Stress Test of Candidate Analyses for the Biomarker-by-Drug Interaction in the Hybrid Design: Exploratory Results

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Purpose. This document reports the results of the stress test set out in the working protocol `docs/46`. Fourteen analyses of the biomarker-by-drug interaction $\beta_{bm:D}$ in the Hybrid design were fitted to the same 28,000 simulated trials in 41 cells, and judged for size, robustness to a prognostic biomarker, sensitivity and power. The study is exploratory: its criteria and its set of analyses were revised as results came in (`docs/46`, Amendments 1 and 2), so the selected analysis is a recommendation to be confirmed in an independent study. The reasoning behind several revisions is in `docs/48`; the recommendation for practice is in `docs/49`.

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here or in `docs/46`.

Table 1: Notation index and glossary of symbols used in this document

Symbol	Meaning	Status
Y_{it}	symptom score of participant i at visit t (code `Sx`); $t = 0$ is baseline	canonical
B_i, b_i	biomarker; standardized biomarker	canonical
B_i^c	biomarker centered at the trial's baseline sample mean, $B_i - \bar{B}$, not scaled (code `bmc`, the form the analyses use; see the note below the table)	local (as `docs/56`)
$D_{bc,it}$	exposure-decayed drug indicator, 0 before any exposure (code `Dbc`)	canonical
η_t	expectancy: 1 open label, 0.5 blinded, 0 at baseline (code `De`)	canonical
$t_{1/2}$, $t_{1/2}^a$	true carryover half-life; half-life assumed by the analysis	canonical; local
c_{bm}	biomarker-BR coupling, the predictive effect	canonical
$c_{bm,TV}$, $c_{bm,PB}$, $c_{bm,BL}$	biomarker correlations with the natural course, the placebo response and baseline severity (prognostic couplings)	local
β_{bm}^{TV}	mean moderation of the natural course by b_i (growing prognostic pathway)	canonical
$\beta_{bm:D}$	biomarker-by-drug interaction, the estimand	canonical
ρ	AR(1) serial correlation per week	canonical
N	total sample size across paths	canonical
α	nominal level, 0.05	canonical
π, $\hat{\pi}$	power; an estimated rejection rate (size or power)	canonical
n_{reps}	Monte Carlo replicates per cell	canonical

(Corrected 2026-10-08: `bmc` was described here as the standardized biomarker. In `11-strawman-stress-test.R` it is $B_i - \bar{B}$, centered at the trial's baseline sample mean and not divided by an SD. Test statistics and p-values are invariant to rescaling the biomarker, so no reported rate changes; the fitted $\beta_{bm:D}$ is per unit of the raw biomarker. The terms `bmc:De` are therefore written $B_i^c\eta_t = (B_i - \bar{B})\eta_t$ below (renamed from $b_i\eta_t$ in the notation pass of 2026-10-09). See `analysis/report/compendium-mathematics.tex`, Section "The stress-test construct and its fourteen analyses".)

The fourteen analyses (`docs/46`, Section 4 and Amendment 2):

Table 2: The fourteen analyses compared in the stress test

ID	Analysis
S	random intercept plus continuous-time AR(1) residuals (`corCAR1`), CR2 standard errors with Satterthwaite degrees of freedom, baseline as a response row, linear time, $D_{bc,it}$ at $t^a_{1/2} = 0.5$ week
DS	S with $t^a_{1/2}$ chosen by AIC over {0, 0.25, 0.5, 1, 2}
C1, C2	S with residual variances by phase (baseline, open label, blinded); CR2 (C1) or model-based (C2) inference
F1	S plus the expectancy terms η_t and $B^c_i\eta_t$ (`De + bmc:De`)
F1A	F1 with $t^a_{1/2}$ chosen by AIC
G1, G2	F1 with residual variances by phase; CR2 (G1) or model-based (G2) inference
F2	S plus `bmc:t`
ES, EA	S and A with the biomarker's baseline association freed (`+ bmc:bl0`)
A	unstructured covariance with Kenward-Roger inference, otherwise as S
DA	A with $t^a_{1/2}$ chosen by AIC
R	the published reference arm: compound-symmetry covariance (the `cs` structure of `mmrm`, by visit within participant, equivalent to a random intercept with independent errors), model-based standard errors with Satterthwaite degrees of freedom, binary drug indicator (wording corrected 2026-10-08; `docs/46` described it as a random intercept with model-based inference)

Glossary.

- **Null and alternative cells.** A null cell has $c_{bm} = 0$ (1,000 replicates); its alternative twin has $c_{bm} = 0.30$, or 0.25 under compound symmetry (500 replicates).
- **Prognostic cell.** A cell in which the biomarker is associated with the outcome regardless of treatment: through the natural course (constant, $c_{bm,TV} = 0.3$, or growing, $\beta^{TV}_{bm} = 0.3$), the placebo response ($c_{bm,PB} = 0.3$) or baseline severity ($c_{bm,BL} = 0.3$).
- **Size-adjusted power.** Power when each analysis rejects at the 5% quantile of its own p-values in the matching null cell.
- **Shortfall.** In an alternative cell, the gap between an analysis's size-adjusted power and that of the best nominal analysis; the *maximin shortfall* of an analysis is its largest shortfall over the cells considered. (Clarified 2026-10-08: the term is kept because the decision outputs use it; selecting the analysis with the smallest largest shortfall is a minimax-regret rule. The best analysis in each cell is taken over all survivors of the size and sensitivity screens, including the flagged C1 and C2, so shortfalls are measured against the phase-variance models. See

`analysis/report/compendium-mathematics.tex`, Section "Size, power screens and the selection criteria of the stress test".)

- **MCSE.** Monte Carlo standard error, $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$ for a rejection rate $\hat{\pi}$ over n_{reps} replicates.

1. Summary

All results are from stored simulation outputs (verified; Section 11).

1. **Size.** Ten of the fourteen analyses hold their size over the 11 non-prognostic null cells (pooled 0.047 to 0.053, within the 2-MCSE band 0.046 to 0.054, and no cell above 0.075). Four fail: the unstructured model A (0.057 pooled, 0.087 at $N = 35$), its AIC version DA (0.065), the published reference arm R (0.069), and F1A (0.057, and 0.073 when $t_{1/2} = 1$ week).
2. **A prognostic biomarker is the decisive stress.** Every analysis that shares the biomarker's main effect between the baseline row and the later visits rejects too often in at least one prognostic null cell: S 0.100, C1 0.382, A 0.416. Five analyses stay within the cap in all four prognostic null cells: ES, EA, F1, G1 and G2.
3. **Sensitivity.** Every analysis has monotone size-adjusted power curves in both constructs. The symmetry screen was dropped, because power is symmetric in the sign of c_{bm} by construction.
4. **Power.** Robustness costs power. At $c_{bm} = 0.30$ under AR(1), size-adjusted power is 0.514 for S and 0.364 for F1, 29% less; under compound symmetry at $c_{bm} = 0.25$ the difference is small (0.554 against 0.522). F1 has significantly lower power than DS in 15 of the 22 non-prognostic alternative cells (McNemar test at 0.05) and higher power in none.
5. **Selection.** Among the five robust analyses the maximin shortfall over the non-prognostic alternative cells is G2 0.232, F1 0.242, G1 0.252, ES 0.278 and EA 0.318. G2 and F1 lie within the tie margin of 0.02, neither reaches power 0.80 on the grid, and the rule's last tie-break (fewer covariance parameters) selects **F1**. G1 and G2 are effectively equivalent alternatives.
6. **A second reading.** Over all 26 alternative cells, prognostic cells included, G1 (0.308) and G2 (0.312) lead F1 (0.358) by about 0.05. The protocol does not say which set of cells governs the selection, so both readings are reported (Section 7).

2. Design in brief

The design is set out in `docs/46`; this section records what was run.

- **Design.** Hybrid, visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20 plus baseline, four paths with the published allocation.

- **Data-generating process.** Configuration C of `docs/36`: separable AR(1) correlation at $\rho = 0.7$, graded coupling of the biomarker to the drug response, the published recursive carryover on the drug-response mean, the published Gompertz means and SDs, the fixed published allocation, and a stop on any matrix that is not positive definite; base cell $N = 70$, $t_{1/2} = 0.5$ week, no dropout, exponential decay. (Corrected 2026-10-08: this bullet said "anchored carryover". `11-strawman-stress-test.R` decays the previous visit's mean with the cumulative time since discontinuation, the published recursion (inspected). The construct is therefore the covariance construct of the refined process of paper 01, not the refined process in full.)
- **Cells.** 41: two power curves (AR(1), c_{bm} from -0.30 to 0.45; compound symmetry, -0.20 to 0.25) and stress cells varying one factor each (measurement error, $N = 35$ and 150, $t_{1/2} = 0$ and 1 week, Weibull decay with shape 0.5 and 2, 20% MCAR and MAR dropout, and the four prognostic pathways). There are 15 null and 26 alternative cells and 28,000 simulated trials.
- **Analyses.** All fourteen are fitted to the same trials, so every comparison is paired.
- **How the protocol changed** (`docs/46`, Amendments 1 and 2):
    1. the reference analysis became DS, and the symmetry screen was dropped;
    2. the two baseline-severity cells (40 and 41) were added and judged with the other prognostic cells;
    3. G1, G2 and F1A were added and judged by the same rule;
    4. failed fits are excluded from rejection rates, rather than counted as non-rejections as first written; the two conventions differ by at most 0.0025 in any rate.

3. Size

Table 3: Null rejection rates by analysis: pooled over the 11 non-prognostic null cells, largest single cell, and the four prognostic null cells

Analysis	Pooled	Largest cell	Nominal	TV, constant	TV, growing	Placebo	Baseline	Prognostic flag
S	0.048	0.059 (MCAR dropout)	yes	0.099	0.049	0.100	0.050	yes
DS	0.053	0.068 (MCAR dropout)	yes	0.115	0.058	0.112	0.055	yes
C1	0.053	0.069 ($N = 35$)	yes	0.382	0.173	0.205	0.054	yes
C2	0.052	0.066 ($N = 35$)	yes	0.393	0.172	0.193	0.050	yes

Analysis	Pooled	Largest cell	Nominal	TV, constant	TV, growing	Placebo	Baseline	Prognostic flag
F1	0.051	0.062 (AR(1) curve)	yes	0.047	0.046	0.040	0.040	no
F1A	0.057	0.073 ($t_{1/2} = 1$)	no	0.059	0.052	0.047	0.045	no
G1	0.049	0.060 (AR(1) curve)	yes	0.055	0.044	0.042	0.042	no
G2	0.047	0.066 (AR(1) curve)	yes	0.062	0.043	0.045	0.046	no
F2	0.049	0.059 (MCAR dropout)	yes	0.090	0.048	0.098	0.048	yes
ES	0.048	0.059 (AR(1) curve)	yes	0.049	0.063	0.049	0.050	no
EA	0.052	0.070 ($N = 35$)	yes	0.049	0.057	0.050	0.066	no
A	0.057	0.087 ($N = 35$)	no	0.416	0.195	0.222	0.051	yes
DA	0.065	0.103 ($N = 35$)	no	0.509	0.245	0.256	0.064	yes
R	0.069	0.081 ($N = 35$)	no	0.107	0.053	0.159	0.075	yes

The pooled rate is over 11,000 replicates, so its 2-MCSE band is 0.046 to 0.054; a single cell has 1,000 replicates (MCSE 0.0069), and the criterion caps each cell at 0.075. An analysis is flagged when any prognostic null cell exceeds 0.075.

Reading.

- **Inference matters most at small N.** The unstructured model with Kenward-Roger inference [5] rejects 0.087 at $N = 35$, and with an AIC-chosen half-life 0.103. The random-intercept models with CR2 standard errors [3, 4] stay at or below 0.069 in every cell, except F1A (0.073), whose half-life is chosen by AIC.
- **Choosing the half-life by AIC inflates size slightly.** DS (0.053) passes; F1A (0.057) and DA (0.065) do not. The selection step adds variability that the standard errors do not reflect.
- **The published reference arm is liberal** throughout (0.069 pooled), as earlier documents found (`docs/45`).
- **Baseline severity alone does little harm.** A prognostic association with baseline severity enters every visit equally, so every analysis stays within the cap in that cell (0.040 to 0.075; R reaches the cap exactly, and EA reaches 0.066).

4. Robustness to a prognostic biomarker

Analyses that keep baseline as a response row and use one biomarker main effect for it and for the later visits attribute part of a prognostic association to the

interaction (derived, `docs/48`, Section 2). The size table shows this directly: S, DS, C1, C2, F2, A, DA and R are flagged, with null rejection rates up to 0.509 when the natural course is associated with the biomarker. F2's `bmc:t` term remedies the growing natural-course pathway (0.048) but not the constant one (0.090) or the placebo pathway (0.098).

Two families stay within the cap in every prognostic cell:

- **the expectancy-term models** F1, G1 and G2, which add η_t and $B_i^c \eta_t$ and so let the biomarker's association differ between baseline, open-label and blinded visits;
- **the free-baseline models** ES and EA, which give the baseline row its own biomarker slope.

Staying within the cap is an empirical result, not exact invariance. In the Hybrid design F1 leaves a small residual bias under a constant natural-course pathway, because the blinded on-drug and off-drug visits carry unequal weight, and ES leaks a little under the placebo pathway through the open-label visits (derived; `docs/49`, Section 3, and `docs/48`, Section 2).

The protocol's rule recommends a flagged analysis only together with a remedy (ES, EA, F1 or F2) that is nominal in the failing cell. The remedies that pass every prognostic cell are themselves candidates, so the selection is made among the unflagged analyses.

5. Sensitivity

Table 4: Size-adjusted power curves under AR(1): monotonicity in both constructs, power at plus and minus 0.30, steepness between 0.15 and 0.30, and the smallest coupling reaching power 0.80

Analysis	Monotone (AR(1) / CS)	Power at $c_{bm} = -0.30$ / $+0.30$	Steepness	Detectable coupling
S	yes / yes	0.450 / 0.514	2.32	0.446
DS	yes / yes	0.422 / 0.486	2.21	not reached
C1	yes / yes	0.524 / 0.578	2.49	0.403
C2	yes / yes	0.554 / 0.606	2.65	0.403
F1	yes / yes	0.352 / 0.364	1.80	not reached
F1A	yes / yes	0.326 / 0.338	1.67	not reached
G1	yes / yes	0.340 / 0.354	1.72	not reached
G2	yes / yes	0.360 / 0.380	1.75	not reached
F2	yes / yes	0.446 / 0.516	2.36	not reached
ES	yes / yes	0.336 / 0.364	1.71	not reached

Analysis	Monotone (AR(1) / CS)	Power at $c_{bm} = -0.30$ / $+0.30$	Steepness	Detectable coupling
EA	yes / yes	0.300 / 0.306	1.48	not reached
A	yes / yes	0.510 / 0.534	2.48	0.430
DA	yes / yes	0.472 / 0.498	2.25	0.434
R	yes / yes	0.394 / 0.398	1.76	not reached

Reading.

- **Every curve is monotone**, so the sensitivity screen removes no analysis.
- **The symmetry screen was defective.** The data-generating process treats $+c_{bm}$ and $-c_{bm}$ alike, so power is symmetric in theory; the observed differences between -0.30 and $+0.30$ (at most 0.070, F2, against an MCSE of about 0.03 for a difference of two independent cells of 500) are consistent with Monte Carlo error, and the screen would have eliminated analyses for noise. It was dropped in Amendment 1 (derived, `docs/48`, Section 4).
- **The robust analyses are flatter.** Their steepness is 1.5 to 1.8, against 2.2 to 2.7 for the flagged analyses other than R (1.76), and none reaches power 0.80 by $c_{bm} = 0.45$ at $N = 70$. Steepness and detectable coupling therefore cannot break ties among them.

6. Power

Table 5: Size-adjusted power (unadjusted in parentheses) in the two reference alternative cells, $N = 70$

Analysis	AR(1), $c_{bm} = 0.30$	Compound symmetry, $c_{bm} = 0.25$
S	0.514 (0.508)	0.554 (0.554)
DS	0.486 (0.516)	0.480 (0.476)
C1	0.578 (0.602)	0.660 (0.658)
C2	0.606 (0.606)	0.656 (0.630)
F1	0.364 (0.410)	0.522 (0.544)
F1A	0.338 (0.436)	0.446 (0.508)
G1	0.354 (0.408)	0.546 (0.552)
G2	0.380 (0.426)	0.528 (0.504)
F2	0.516 (0.496)	0.552 (0.552)
ES	0.364 (0.412)	0.662 (0.634)
EA	0.306 (0.392)	0.602 (0.596)
A	0.534 (0.566)	0.618 (0.618)
DA	0.498 (0.592)	0.572 (0.596)
R	0.398 (0.476)	0.596 (0.594)

Analysis	AR(1), $c_{bm} = 0.30$	Compound symmetry, $c_{bm} = 0.25$

Table 6: Size-adjusted power at $c_{bm} = 0.30$ in the stress cells: the reference analysis S, the most powerful analysis C1, and the five robust analyses

Cell	S	C1	F1	G1	G2	ES	EA
$N = 150$	0.828	0.918	0.748	0.752	0.754	0.704	0.698
$N = 35$	0.222	0.272	0.192	0.214	0.218	0.206	0.196
$t_{1/2} = 0$	0.594	0.666	0.496	0.510	0.532	0.504	0.490
$t_{1/2} = 1$ week	0.388	0.502	0.272	0.282	0.306	0.288	0.294
Weibull decay, shape 0.5	0.404	0.514	0.308	0.296	0.378	0.270	0.230
Weibull decay, shape 2	0.588	0.682	0.504	0.526	0.526	0.458	0.464
Measurement error	0.400	0.444	0.318	0.320	0.330	0.336	0.324
20% MCAR dropout	0.426	0.542	0.378	0.394	0.406	0.360	0.346
20% MAR dropout	0.418	0.518	0.382	0.404	0.378	0.356	0.322
Prognostic, TV constant	0.606	0.774	0.482	0.520	0.524	0.370	0.378
Prognostic, TV growing	0.516	0.672	0.378	0.484	0.490	0.192	0.208
Prognostic, placebo	0.652	0.736	0.446	0.428	0.424	0.498	0.438
Prognostic, baseline	0.444	0.552	0.396	0.394	0.406	0.336	0.290

Reading.

- **The phase-variance models are the most powerful** (C1 has the highest power in every stress cell shown), but they fail the prognostic cells badly, so their power is not available to a trial whose biomarker may be prognostic.
- **The cost of robustness depends on the correlation structure.** Under AR(1) the robust analyses have 26% (G2) to 40% (EA) less size-adjusted power than S at $c_{bm} = 0.30$; under compound symmetry ES even exceeds S (0.662 against 0.554), and F1 is close (0.522).
- **The power reported for flagged analyses in prognostic cells is size-adjusted**, so it is not the power a trial would see: unadjusted, those analyses reject far too often under the null.
- **The free-baseline models lose most when the natural course grows with the biomarker** (0.192 and 0.208), which is why their all-cells shortfall is large (Section 7).

7. Selection

The rule (`docs/46`, Section 7, as revised by Amendment 1) is applied to the analyses that pass the size screen and the sensitivity screen and are not flagged in a prognostic cell: ES, EA, F1, G1 and G2.

Table 7: Maximin shortfall in size-adjusted power among the analyses that pass the size and sensitivity screens, over the non-prognostic alternative cells and over all alternative cells

Analysis	Largest shortfall, non-prognostic cells	Worst cell	Largest shortfall, all cells	Worst cell
C1 (flagged)	0.034	Weibull 0.5	0.064	TV growing
C2 (flagged)	0.040	Weibull 2	0.040	Weibull 2
S (flagged)	0.144	Weibull 0.5	0.220	TV growing
F2 (flagged)	0.160	Weibull 0.5	0.244	TV growing
DS (flagged)	0.182	CS curve, 0.25	0.214	TV growing
G2	0.232	AR(1) curve, 0.375	0.312	placebo
F1	0.242	AR(1) curve, 0.30	0.358	TV growing
G1	0.252	Weibull 0.5	0.308	placebo
ES	0.278	Weibull 0.5	0.544	TV growing
EA	0.318	Weibull 0.5	0.528	TV growing

Outcome.

1. **Smallest shortfall.** Over the non-prognostic alternative cells, G2 (0.232) and F1 (0.242) are the two best robust analyses.
2. **Tie.** They differ by 0.010, inside the rule's tie margin of 0.02.
3. **First tie-break.** The smaller detectable coupling in the base construct is undefined: neither reaches power 0.80 on the grid.
4. **Last tie-break.** The analysis with fewer covariance parameters, **F1**, is selected. F1 is also the only one of the two with CR2 inference; G2 relies on model-based standard errors.
5. **Strawman rule.** The reference analysis DS is flagged in the prognostic cells, so it cannot be retained; F1 replaces it. The paired comparison makes the price explicit: F1 has significantly lower size-adjusted power than DS in 15 of the 22 non-prognostic alternative cells, and higher power in none.

The all-cells reading. If the prognostic alternative cells are included in the maximin set, the order among the robust analyses is G1 0.308, G2 0.312, F1 0.358, EA 0.528, ES 0.544, and the phase-variance versions lead F1 by about 0.05. The working protocol does not say whether those cells count. The non-prognostic reading is reported as the outcome because the prognostic cells are

judged by their own screen. Under the all-cells reading G1 and G2 would tie (0.004 apart) ahead of F1, and the tie-breaks could not separate them, since they share one model and differ only in inference. The difference from F1 arises mainly when the natural course is associated with the biomarker (G1 0.484 against F1 0.378 in that cell).

8. Choosing the half-life by AIC

DS, DA and F1A choose $t^a_{1/2}$ by AIC from {0, 0.25, 0.5, 1, 2} weeks (0 meaning a binary drug indicator), with maximum likelihood for the choice and a REML refit.

Table 8: Share of fits choosing each analysis half-life, over all cells

Analysis	0	0.25	0.5	1	2
DS	0.170	0.144	0.299	0.208	0.179
DA	0.188	0.144	0.294	0.181	0.193
F1A	0.236	0.163	0.292	0.190	0.120

The choice is spread across the grid even though the true half-life is 0.5 week in most cells, and the selection step costs size: F1A fails the size criterion where F1 passes, and DS loses about 0.03 of size-adjusted power against S at the AR(1) reference cell (0.486 against 0.514). A fixed $t^a_{1/2} = 0.5$ week is therefore preferred with F1.

9. Failed fits

A fit that returned an error was recorded as missing. Failures number 0 to 12 per analysis over 28,000 trials (G1 and G2 12; S, C1 and C2 6; F1, F1A and ES 3; DS and F2 2; A, DA, EA and R 0), and at most 0.6% in any cell (G1 and G2), below the 2% at which an analysis would be flagged.

10. Limitations

- **Exploratory.** The criteria, the reference analysis and the set of analyses were revised after results were seen. The selection is therefore a recommendation from an exploratory comparison, and should be confirmed in an independent simulation study before it is relied on.
- **One design and one effect size.** Only the Hybrid design was studied, with the alternative cells at $c_{bm} = 0.30$ (0.25 under compound symmetry) and prognostic couplings of 0.3.

- **The set of cells governs the maximin.** The outcome depends on whether the prognostic alternative cells count (Section 7).
- **Monte Carlo error.** Alternative cells have 500 replicates (MCSE of power at most 0.022), so differences in shortfall below about 0.03 between two analyses are within noise for a single cell.
- **The prognostic mechanisms exist only in the stress-test driver,** not yet in the package (`docs/44`).

11. Reproducibility

All numbers in this document were read from stored outputs on 2026-10-08:

- `analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R` produced the 28,000 trials and the fits (main run, `--extra` for G1 and G2, `--extra2` for F1A); replicate and summary files are in `analysis/data/quick-sim/carryover-closed-form/strawman-stress/` (`replicates-full*.rds`, `summary-full*.csv`).
- `12-strawman-stress-decision.R --strawman DS --no-symmetry --with-extras` applies the rule to all fourteen analyses and writes `decision-screen-ds-nosym-all14.csv`, `decision-sensitivity-ds-nosym-all14.csv`, `decision-power-ds-nosym-all14.csv` and `decision-maximin-ds-nosym-all14.csv`.
- Seeds are set per chunk as $20261007 + 1000 \times \text{cell} + \text{chunk}$, so results do not depend on the number of cores.

12. References

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